Unit 2 Homework — Exponent Rules — Answer Key

Complete independently · Exponent Rules

A. Exponent Rules — Product & Quotient

Problem 1 Score 3

Simplify: $x^5 \cdot x^6$

Answer: x^{11}

Problem 2 Score 3

Simplify: $b^9 \div b^4$

Answer: b^5

Problem 3 Score 4

Simplify: $(4x^2)(2x^5)$

Answer: $8x^7$

Problem 4 Score 4

Simplify: $24m^8 \div 6m^3$

Answer: $4m^5$

Problem 5 Score 4

Simplify: $c^3 \cdot c^4 \cdot c^2$

Answer: c^9

Problem 6 Score 4

Simplify: $(7y^4)(2y)$

Answer: $14y^5$

B. Power, Zero & Negative Exponents

Problem 1 Score 3

Simplify: $(y^5)^2$

Answer: y^{10}

Problem 2 Score 3

Simplify: 12^0

Answer: 1

Problem 3 Score 4

Simplify: $(3x^2)^2$

Answer: $9x^4$

Problem 4 Score 4

Simplify: $(8m^3n^2)^0$

Answer: 1

Problem 5 Score 4

Simplify: 2^{-3} (write as a fraction)

Answer: $\frac{1}{8}$

Problem 6 Score 4

Simplify: y^{-2} (write as a fraction)

Answer: $\frac{1}{y^2}$

Problem 7 Score 5

Evaluate -2^4 (no parentheses).

Answer: -16

Problem 11 Score 5

Lena says $2^{-3} = -8$. Which fixes her mistake?

A. Flip the base: $(1/2)^3 = 1/6$

B. A negative exponent means a reciprocal: $2^{-3} = 1/2^3 = 1/8$

C. Lena is right. A negative exponent makes the answer negative.

D. Multiply instead: $2 \times (-3) = -6$

Answer: B

Problem 12 Score 5

Simplify: $(3x^2y^3)^2 \cdot (2x^{-1}y^4)$

A. $18x^5y^{10}$

B. $18x^3y^9$

C. $18x^3y^{10}$

D. $6x^3y^{10}$

Answer: C

Problem 13 Score 5

Simplify: $(6a^4b)(2a^{-1}b^3) \div (4a^3b^6)$

A. $8/b^2$

B. $3a^6/b^2$

C. $3/b^2$

D. $3b^2$

Answer: C